Forcing Graphs and Cyclic Completion Energy in Two-Comparable Triple Systems

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Abstract

Let $G(n)$ be the maximum size of a subset of $[n]^3$ in which every two distinct triples are strictly ordered in at least two coordinates. The best known general upper bound is $o(n^2)$, and it remains open whether $G(n) = O(n^{2-\delta})$ for an absolute $\delta > 0$. We study the repeated-coordinate constraints behind this problem through its equivalent formulation by monotone partial matrices. Every pair of equal-label entries forces two cross holes. We orient these holes canonically and obtain an edge-coloured bipartite forcing graph: vertex degrees are forcing multiplicities, neighbourhoods are strict product-order chains, colour classes are order-preserving matchings, and every four-cycle is rainbow. In the three coordinate representations, the second completion energies satisfy the cyclic packing law

$$E_x + E_y + E_z \le \binom{|M|}{2}.$$

An exact decomposition of the unused capacity records coordinate-degree variance, forcing-degree variance, aligned pairs, and unsaturated inversions. We also prove endpoint-exclusion inequalities for common fibres. Finally, the centres of all pressure grids through a fixed cell split into two signed sets, each admitting two canonical encodings as a new two-comparable set. This gives pointwise bounds $L_0 \le 2|M|$ and $L_1 \le 2|M|^{3/2}$, with the exponent $3/2$ sharp, and yields a fixed power saving under a quantitative pressure-load hypothesis. These results impose simultaneous structural restrictions on a hypothetical near-quadratic family but do not establish such a power saving.

1 Introduction

For $u, v \in [n]^3$, write $u \prec_2 v$ if u is strictly smaller than v in at least two coordinates. A set $M \subseteq [n]^3$ is *pairwise two-comparable* if, for every distinct $u, v \in M$, either $u \prec_2 v$ or $v \prec_2 u$. We consider the extremal function

$$G(n) = \max\{|M| : M \subseteq [n]^3 \text{ is pairwise two-comparable}\}.$$

The projection bound $G(n) \le n^2$ is immediate, since two members of M cannot agree in two coordinates. The problem is nevertheless far from this elementary bound. The related two-increasing sequence problem was posed by Loh [5]. Gowers and Long proved, using quantitative triangle removal, that

$$G(n) \le \frac{n^2}{\exp(\Omega(\log^* n))} = o(n^2),$$

and gave constructions of size at least $n^{1.546}$ for arbitrarily large n [4, 2]. The fixed-power question

$$G(n) = O(n^{2-\delta}) \qquad \text{for some absolute } \delta > 0 \tag{1.1}$$

remains open. Equivalent formulations occur in monotone matrices and tripod packing; see Stein [7], Tiskin [8], Östergård and Pöllänen [6], and Aronov et al. [1]. The distinction from a two-increasing *sequence* is important: Gowers and Long obtained a power saving for sequences by using acyclicity, which is absent for general two-comparable sets. Recent higher-dimensional results for the related increasing-sequence and tournament problems likewise leave the three-coordinate comparable-set problem separate [3].

The present paper develops structural constraints on a two-comparable family. Choose one coordinate as a label and the other two as row and column. The result is a partial matrix in which rows, columns, and equal-label positions are all strictly increasing. If a label occurs in both the row and the column of an empty cell, then that label *forces* the cell to remain empty. The main results organize these forcing events in four ways.

First, Section 3 constructs a canonical oriented forcing graph. It is a simple edge-coloured bipartite graph whose degrees and wedges are the first and second completion counts. Its neighbourhoods are ordered chains, its colour classes are order-preserving matchings, and all four-cycles and bicliques are rainbow. Complete bipartite subgraphs of every fixed size can nevertheless occur, so the structure is not captured by a fixed forbidden subgraph.

Second, Section 4 proves an endpoint-exclusion theorem for any collection of common row fibres and derives a hierarchy of binomial-moment inequalities. Third, Sections 5 and 6 couple all three coordinate representations. A double-forcing event is put in bijection with a saturated inversion. The three inversion images are disjoint, giving

$$E_x + E_y + E_z \le \binom{|M|}{2}. \tag{1.2}$$

The associated exact slack identity retains every loss in the two variance steps and in the cyclic packing step. It gives an elementary density calibration, an equality-stability statement, and a criterion showing that a near-quadratic family must have almost maximal polynomial-scale forcing energy in every coordinate representation.

Fourth, Section 7 studies the overlap of the rectangular pressure grids forced around holes. For a fixed physical cell, the centres of all pressure grids containing it split according to orientation. Each signed centre set has two canonical encodings as a pairwise two-comparable subset of $[n]^3$. A discrete Loomis–Whitney estimate then gives the pointwise load bounds

$$L_0(z) \le 2|M|, \qquad L_1(z) \le 2|M|^{3/2}. \tag{1.3}$$

The weighted exponent is sharp as a function of $|M|$. Moreover, if the maximum weighted pressure load is $O(n^{3-\eta})$, then

$$|M| = O(n^{2-\eta/6} + n).$$

Thus a concrete overlap estimate would imply (1.1); the unconditional pointwise estimate identifies the overlap as a smaller copy of the original two-comparable problem.

All results in the body of the paper are proved mathematically and do not depend on computation. Appendix A describes an exhaustive order-three verification used to check definitions, factors of two, and the explicit constructions. Section 8 records the precise limits of the methods. In particular, no absolute $\delta > 0$ is obtained here.

2 Monotone partial matrices and completion counts

We first give the matrix correspondence and fix notation. An order-n partial matrix A has cells $[n] \times [n]$, some of which are filled by labels from $[n]$. Its sets of filled cells and holes are denoted by

$\mathcal{F}(A)$ and $\mathcal{H}(A)$. It is *monotone* if:

(i) the filled entries in each row strictly increase from left to right;

(ii) the filled entries in each column strictly increase from top to bottom;

(iii) the cells carrying any fixed label form a strict chain in the row–column product order.

Here $(i,j) < (i',j')$ means $i < i'$ and $j < j'$.

Given a pairwise two-comparable set $M \subseteq [n]^3$, choose a coordinate $q \in \{x,y,z\}$ as the label and use the remaining two coordinates, in their cyclic order, as row and column. Since two triples cannot agree in two coordinates, this defines a partial matrix $A^{(q)}$. A pair agreeing in the row, column, or label coordinate gives, respectively, conditions (i), (ii), or (iii). Conversely, these three conditions guarantee two-comparability for pairs agreeing in one coordinate; if all three coordinates differ, one of the two triples is smaller in at least two coordinates by the majority principle. We have therefore proved the following standard equivalence.

Proposition 2.1 (Matrix correspondence)**.** *Pairwise two-comparable subsets of* $[n]^3$ *are in bijection with order-*n *monotone partial matrices. The bijection may use any coordinate as label.*

Fix one representation A. If a hole $h = (i,j)$ has a label k occurring both in row i and in column j, then k *forces* h. Write

$$K_h = \{k : k \text{ forces } h\}, \qquad d(h) = |K_h|.$$

The forcing multiplicity is zero for an unforced hole. Let t_k be the number of occurrences of label k. The first completion count is exact.

Lemma 2.2 (First completion identity)**.** *For every monotone partial matrix,*

$$\sum_{h \in \mathcal{H}(A)} d(h) = \sum_{k=1}^{n} t_k(t_k - 1). \tag{2.1}$$

Proof. Order the t_k occurrences of a fixed label k as $(r_1, c_1) < \cdots < (r_{t_k}, c_{t_k})$. For every $p < q$, the two cross cells (r_p, c_q) and (r_q, c_p) are holes: filling either would contradict row and column monotonicity. Both are forced by k. Conversely, a forcing event (h,k) uniquely determines the occurrence of k in the row of h and the occurrence of k in its column, hence one of these two cross cells. Thus label k contributes $2\binom{t_k}{2} = t_k(t_k-1)$ events. Summing over k proves the identity. $\square$

The *second completion energy* is

$$E(A) = \sum_{h \in \mathcal{H}(A)} \binom{d(h)}{2}, \tag{2.2}$$

where $\binom{d}{2} = 0$ for $d \in \{0,1\}$. It counts a hole together with an unordered pair of labels that both force it.

3 The oriented forcing graph

We begin with the local orientation that makes the graph canonical.

Lemma 3.1 (Common orientation). *Let $h = (i, j)$ be forced by a label k, whose occurrences in row i and column j are (i, c) and (r, j). Then $c < j$ if and only if $i < r$. Moreover, all labels forcing the same hole have the same one of the two orientations*

$$c < j < \text{ and } i < r, \qquad \text{or} \qquad j < c \text{ and } r < i.$$

Proof. The two occurrences of k are comparable in the strict product order, so $c < j$ holds exactly when $i < r$. Now let $a < b$ force h. Their positions in row i increase with the labels, and their positions in column j do so as well. If a were in the first orientation and b in the second, their row positions would have the correct order but their column positions would have the reverse order. If the orientations were reversed, the contradiction would occur in the row. Hence the orientation is common. □

Call the first orientation *left-facing* and the second *right-facing*; let $\mathcal{H}_L$ and $\mathcal{H}_R$ be the corresponding sets of forced holes. For the ordered occurrences $(r_1, c_1) < \cdots < (r_{t_k}, c_{t_k})$ of label k, put

$$\ell^k_{pq} = (r_p, c_q), \qquad r^k_{pq} = (r_q, c_p) \qquad (p < q).$$

The first cross hole lies in $\mathcal{H}_L$ and the second in $\mathcal{H}_R$.

Definition 3.2 (Forcing graph). The *forcing graph* $\Gamma(A)$ is the edge-coloured bipartite graph with parts $\mathcal{H}_L$ and $\mathcal{H}_R$. For every label k and $p < q$, it has an edge of colour k joining ℓ^k_{pq} to r^k_{pq}.

A matching between subsets of two product-ordered grids is *order-preserving* if $x < x'$ for two first endpoints implies $y < y'$ for their corresponding second endpoints.

Theorem 3.3 (Forcing-graph normal form). *For every monotone partial matrix A, the graph $\Gamma(A)$ has the following properties.*

(a) It is simple and bipartite with parts $\mathcal{H}_L, \mathcal{H}_R$.

(b) The degree of every forced hole h is $d(h)$.

(c) The neighbours of each vertex form a strict product-order chain; in that chain order, the incident edge colours strictly increase.

(d) Each colour class is an order-preserving matching.

(e) Every four-cycle has four distinct edge colours.

(f) Its edge and wedge counts are

$$e(\Gamma(A)) = \sum_{k=1}^{n} \binom{t_k}{2} = \frac{1}{2} \sum_{h \in \mathcal{H}(A)} d(h), \qquad \sum_{h \in V(\Gamma(A))} \binom{\deg h}{2} = E(A). \tag{3.1}$$

Proof. The graph is bipartite by construction. Its two cross holes determine the two opposite filled corners; those corners have the edge colour as their common label. Thus two colours cannot give the same edge, and the graph is simple.

Fix a left-facing hole $h = (i, j)$. A forcing label k occurs at unique cells (i, c_k) and (r_k, j) with $c_k < j$ and $i < r_k$. The edge produced by these two occurrences joins h to (r_k, c_k). Conversely, every incident edge arises in this way, and different labels give different neighbours. Hence $\deg h = d(h)$. If $a < b$ both force h, row and column monotonicity give $c_a < c_b$ and $r_a < r_b$. Therefore the neighbours, and their incident colours, increase together. The right-facing case is identical after reversing both inequalities.

For a fixed label k, each of the maps $(p, q) \mapsto \ell^k_{pq}$ and $(p, q) \mapsto r^k_{pq}$ is injective, so its colour class is a matching. If $\ell^k_{pq} < \ell^k_{uv}$, then $p < u$ and $q < v$, and consequently $r^k_{pq} < r^k_{uv}$. The same argument with the order reversed proves that the matching is order-preserving.

Consider a four-cycle. Its two left vertices are comparable because both belong to the neighbourhood chain of either right vertex; similarly, its two right vertices are comparable. Write them as $x < x'$ and $y < y'$. If the colours at $(x, y), (x, y'), (x', y), (x', y')$ are a, b, c, d, respectively, the neighbourhood order gives

$$a < b < d, \qquad a < c < d.$$

Adjacent edges have different colours because each colour class is a matching. The only possible repetition is $b = c$, but then that colour has endpoints ordered as $x < x'$ on the left and $y' > y$ on the right, contrary to the order-preserving property. Thus the cycle is rainbow.

There is one edge for each unordered pair of occurrences of each label. This proves the first edge equality. Lemma 2.2, or the degree identity and the handshake lemma, gives the second. Finally, summing $\binom{\deg h}{2} = \binom{d(h)}{2}$ over the forced holes gives the wedge identity. □

The rainbow property extends from four-cycles to complete bipartite subgraphs.

Corollary 3.4 (Rainbow bicliques)**.** *Every $K_{s,t}$ contained in $\Gamma(A)$ is rainbow. Consequently $st \le n$. In particular, two vertices in the same part have at most $\lfloor n/2 \rfloor$ common neighbours.*

Proof. Adjacent edges have distinct colours. Two disjoint edges of a biclique are opposite edges of a four-cycle and hence also have distinct colours by Theorem 3.3(e). Thus the st edges have distinct colours, while only n labels are available. The final assertion is the case $s = 2$. □

The next construction shows that no fixed biclique can be excluded.

Proposition 3.5 (Arbitrary forcing bicliques)**.** *For every $s, t \ge 2$, there is an order-n monotone partial matrix with $n = st$ whose forcing graph contains $K_{s,t}$.*

Proof. For $a \in [s]$ and $b \in [t]$, set $\lambda_{ab} = a + s(b - 1)$. Fill two disjoint rectangles by

$$A(a, b) = \lambda_{ab}, \qquad A(s + b, t + a) = \lambda_{ab},$$

and leave all other cells empty. Both rectangles fit because $s + t \le st$. Rows and columns increase, and the two occurrences of every label are strictly ordered in both coordinates; hence the matrix is monotone.

For $a \in [s]$ put $x_a = (a, t + a)$, and for $b \in [t]$ put $y_b = (s + b, b)$. The two filled cells carrying λ_{ab} are the other corners of the rectangle with corners x_a and y_b. Therefore $x_a y_b$ is an edge of colour λ_{ab} for every (a, b), giving the required biclique. □

4 Common fibres and moment inequalities

The forcing constraints also give exact inequalities for common fibres. For a set I of rows, define

$$C(I) = \{j : \text{column } j \text{ is filled in every row of } I\},$$
$$L(I) = \{k : \text{label } k \text{ occurs in every row of } I\}.$$

Let r_i denote the number of filled cells in row i.

Theorem 4.1 (Endpoint exclusion)**.** *For every set I of $p \geq 2$ rows of an order-n monotone partial matrix,*

$$p|L(I)| + |C(I)| \leq n, \qquad |L(I)| + p|C(I)| \leq n, \qquad |L(I)| + |C(I)| \leq \min_{i \in I} r_i. \tag{4.1}$$

The analogous statements obtained by permuting the three coordinate roles also hold.

Proof. For $i \in I$ and $k \in L(I)$, let $c_i(k)$ be the column of the k-cell in row i. We claim that the $p|L(I)|$ values $c_i(k)$ are all distinct. Suppose $c_i(k) = c_{i'}(\ell) = j$ for distinct pairs (i,k) and (i',ℓ). Equality is impossible if $i = i'$ or $k = \ell$, so assume $i < i'$. Column monotonicity gives $k < \ell$. Since the k-cells form a strict product-order chain, $c_i(k) < c_{i'}(k)$; row monotonicity in row i' gives $c_{i'}(k) < c_{i'}(\ell)$. Hence $j < c_{i'}(k) < j$, a contradiction.

None of these column values lies in $C(I)$. Indeed, suppose that the k-cell in row i lies in a common column j, and choose $i' \in I \setminus \{i\}$. If $i < i'$, the entry at (i', j) is larger than k, whereas the equal-label chain places the k-cell of row i' strictly to the right of j; row monotonicity gives the opposite comparison. If $i' < i$, both comparisons reverse and again contradict each other. Thus the $p|L(I)|$ values $c_i(k)$ and the $|C(I)|$ common columns are distinct, proving the first inequality.

Interchange the roles of column and label, retaining the row coordinate. Proposition 2.1 gives another monotone partial matrix, and the first inequality becomes $p|C(I)| + |L(I)| \leq n$. Finally, in any row $i \in I$, the cells with labels in $L(I)$ and the cells in columns $C(I)$ are disjoint by the preceding argument. All are filled, proving the third inequality. Coordinate symmetry gives the remaining versions. $\square$

Let c_j and t_k denote column and label degrees in the fixed representation.

Corollary 4.2 (Common-neighbour moments)**.** *For every $2 \leq p \leq n$,*

$$p \sum_{k=1}^{n} \binom{t_k}{p} + \sum_{j=1}^{n} \binom{c_j}{p} \leq n \binom{n}{p}, \tag{4.2}$$

$$\sum_{k=1}^{n} \binom{t_k}{p} + p \sum_{j=1}^{n} \binom{c_j}{p} \leq n \binom{n}{p}, \tag{4.3}$$

$$\sum_{\substack{I \subseteq [n] \\ |I| = p}} \left(|L(I)| + |C(I)|\right) \leq \sum_{\substack{I \subseteq [n] \\ |I| = p}} \min_{i \in I} r_i. \tag{4.4}$$

All cyclic variants hold.

Proof. A label of degree t_k belongs to $L(I)$ for exactly $\binom{t_k}{p}$ choices of I. A column of degree c_j belongs to $C(I)$ for exactly $\binom{c_j}{p}$ choices. Sum the three inequalities in (4.1) over all p-subsets I of the rows. $\square$

For $p = 2$, if two rows share D labels and Q occupied columns, the first two inequalities say $2D + Q \leq n$ and $D + 2Q \leq n$. These are independent of the row degrees and strengthen the disjoint-cell bound $D + Q \leq \min(r_i, r_{i'})$ in complementary directions.

5 Cyclic completion-energy packing

Return to a pairwise two-comparable set $M \subseteq [n]^3$ of size m. For $q \in \{x, y, z\}$, let $A^{(q)}$ be the cyclic representation with coordinate q as label, and define

$$E_q = E(A^{(q)}) = \sum_{h \in \mathcal{H}(A^{(q)})} \binom{d_q(h)}{2}.$$

A *strict inversion* in a monotone partial matrix is an unordered pair of filled cells (i, j) and (r, c) with $i < r$, $j < c$, and with the label at (i, j) larger than the label at (r, c). It is *saturated* if one of its two cross cells is a hole forced by both endpoint labels. Write $\mathrm{Inv}(A)$ and $\mathrm{Sat}(A)$ for the sets of strict and saturated inversions.

Lemma 5.1 (Saturated-inversion correspondence)**.** *For every monotone partial matrix A, there is a bijection between double-forcing events*

$$(h, \{a, b\}), \qquad a < b, \quad a, b \in K_h,$$

and saturated strict inversions. Consequently

$$E(A) = |\,\mathrm{Sat}(A)| \le |\,\mathrm{Inv}(A)|. \tag{5.1}$$

Proof. Let $h = (i, j)$ be forced by $a < b$. In the left-facing orientation, pair the b-cell in row i with the a-cell in column j. The former is northwest of the latter and has the larger label, so the pair is a strict inversion; their other cross cell is h, forced by both labels. In the right-facing orientation, pair the b-cell in column j with the a-cell in row i. This again gives a saturated strict inversion. This defines the forward map.

Conversely, a saturated inversion together with its saturated cross hole recovers the two endpoint labels and hence the double-forcing event. It remains to prove that a strict inversion cannot have both cross holes saturated. Write its endpoints as a northwest cell (p, q) of label b and a southeast cell (r, s) of label a, where $p < r$, $q < s$, and $a < b$. If the northeast cross hole (p, s) is saturated, then row p contains an additional a-cell strictly left of (p, q). If the southwest cross hole (r, q) is saturated, then column q contains an additional a-cell strictly above (p, q). Those two additional a-cells are strictly anti-aligned, contradicting the equal-a chain condition. Hence there is a unique saturated cross hole, so the inverse is well defined. The two constructions are inverse to one another, proving bijectivity. □

Theorem 5.2 (Cyclic energy packing)**.** *For every pairwise two-comparable $M \subseteq [n]^3$ of size m,*

$$\boxed{E_x + E_y + E_z \le \binom{m}{2}.} \tag{5.2}$$

More precisely, the saturated-inversion images in the three coordinate representations are pairwise disjoint sets of unordered pairs from M.

Proof. Apply Lemma 5.1 to the three representations. A strict inversion in $A^{(q)}$ is a pair of triples for which the two non-q coordinates change in the same direction and the q coordinate changes in the opposite direction. Thus q is the unique exceptional coordinate of the pair. The same unordered pair cannot have two distinct unique exceptional coordinates, so the three saturated-inversion images are disjoint. Each image is contained in $\binom{M}{2}$, proving (5.2). □

By Theorem 3.3, E_q is the wedge count of the forcing graph in representation q. Thus the total number of wedges in the three canonical graphs is packed into a single multiplicity-one resource: the unordered pairs of members of M.

6 Exact slack and consequences

Let $t_u^{(q)}$ be the number of members of M whose q coordinate is u, and put

$$S_q = \sum_{u=1}^{n} t_u^{(q)}(t_u^{(q)} - 1). \tag{6.1}$$

Every representation has $H = n^2 - m$ holes. For $n \geq 2$ one has $H > 0$: otherwise Lemma 2.2 would force every label degree to be at most one, contrary to $m = n^2 > n$. Define

$$V_q = \sum_{u=1}^{n} \left(t_u^{(q)} - \frac{m}{n} \right)^2, \qquad W_q = \sum_{h \in \mathcal{H}(A^{(q)})} \left(d_q(h) - \frac{S_q}{H} \right)^2. \tag{6.2}$$

Let $P_\parallel$ count unordered pairs of members of M whose three coordinates are distinct and all change in the same direction. Finally, set

$$U_q = |\operatorname{Inv}(A^{(q)})| - E_q,$$

the number of unsaturated strict inversions in the q representation.

Theorem 6.1 (Exact slack identity)**.** *For every pairwise two-comparable $M \subseteq [n]^3$ of size m and $n \geq 2$,*

$$\boxed{\binom{m}{2} = \frac{1}{2H} \sum_{q \in \{x,y,z\}} S_q^2 + \frac{1}{2} \sum_{q \in \{x,y,z\}} W_q + P_\parallel + \sum_{q \in \{x,y,z\}} U_q.} \tag{6.3}$$

Moreover,

$$S_q = \frac{m^2}{n} - m + V_q. \tag{6.4}$$

All remainder terms $V_q, W_q, P_\parallel, U_q$ are nonnegative.

Proof. Lemma 2.2 gives $\sum_{h \in \mathcal{H}(A^{(q)})} d_q(h) = S_q$. Expanding W_q therefore yields the exact variance identity

$$E_q = \frac{1}{2} \left(\frac{S_q^2}{H} + W_q - S_q \right). \tag{6.5}$$

We next partition the unordered pairs from M. Two distinct members cannot agree in two coordinates, so the number of pairs agreeing in one coordinate is counted exactly once by the coordinate fibres and equals

$$\sum_q \sum_{u=1}^{n} \binom{t_u^{(q)}}{2} = \frac{1}{2} \sum_q S_q.$$

A pair differing in all three coordinates is either counted by $P_\parallel$ or has a unique exceptional coordinate q, in which case it is a strict inversion in $A^{(q)}$. Hence

$$\binom{m}{2} = \frac{1}{2} \sum_q S_q + P_\parallel + \sum_q (E_q + U_q). \tag{6.6}$$

Substituting (6.5) cancels the linear S_q terms and proves (6.3). Finally,

$$V_q = \sum_u (t_u^{(q)})^2 - \frac{m^2}{n},$$

and subtracting $\sum_u t_u^{(q)} = m$ gives (6.4). Nonnegativity follows from the definitions and Lemma 5.1. $\square$

The identity keeps both variance defects and the full unused part of cyclic inversion packing. In particular,

$$S_q \geq \frac{m^2}{n} - m, \qquad E_q \geq \frac{1}{2}\left(\frac{S_q^2}{H} - S_q\right). \tag{6.7}$$

6.1 Density calibration and stability

Corollary 6.2 (Elementary density calibration)**.** *One has*

$$\limsup_{n\to\infty} \frac{G(n)}{n^2} \leq \rho_0 := \frac{\sqrt{13}-1}{6} = 0.434258\ldots.$$

Proof. Consider a sequence with $m/n^2 \to \rho > 0$. The case $\rho = 1$ is impossible: then $H = o(n^2)$ and $S_q \geq m^2/n - m = (1-o(1))n^3$, so (6.7) would give $E_q/n^4 \to \infty$, contrary to Theorem 5.2. Thus $0 < \rho < 1$. For every q,

$$S_q \geq B := \frac{m^2}{n} - m = \Theta(n^3), \qquad H = \Theta(n^2).$$

The function $s \mapsto s^2/H - s$ is increasing on $[B, \infty)$ for all sufficiently large n. It follows from (6.7) that

$$\frac{E_q}{n^4} \geq \frac{\rho^4}{2(1-\rho)} + o(1).$$

On the other hand, Theorem 5.2 gives $\sum_q E_q/n^4 \leq \rho^2/2 + o(1)$. Therefore

$$\frac{3\rho^4}{1-\rho} \leq \rho^2, \qquad \text{equivalently} \qquad 3\rho^2 + \rho - 1 \leq 0.$$

The positive root is ρ_0. ☐

The known bound $G(n) = o(n^2)$ is asymptotically stronger. The corollary is included only as a calibration of the cyclic identity, not as a competitive upper bound.

Corollary 6.3 (Slack stability)**.** *If a sequence of pairwise two-comparable sets satisfies $m/n^2 \to \rho_0$, then, for every $q \in \{x, y, z\}$,*

$$V_q = o(n^3), \qquad W_q = o(n^4), \qquad P_\| + \sum_q U_q = o(n^4).$$

Proof. Put $B = m^2/n - m$, so $S_q = B + V_q$. Subtract $3B^2/(2H)$ from both sides of (6.3). The remainder on the right is the sum of the nonnegative quantities

$$\frac{1}{2H}\sum_q (2BV_q + V_q^2) + \frac{1}{2}\sum_q W_q + P_\| + \sum_q U_q.$$

Since $3\rho_0^2 + \rho_0 - 1 = 0$, the left side is $o(n^4)$. Also $B = \Theta(n^3)$ and $H = \Theta(n^2)$, so each asserted estimate follows. ☐

In view of the known $o(n^2)$ upper bound, the hypothesis of Corollary 6.3 does not describe an extremal sequence. Its role is to record the equality conditions of the elementary cyclic argument.

6.2 A low-energy criterion

Corollary 6.4 (Low-energy power saving)**.** *Fix $\eta > 0$. If $E_q = O(n^{4-\eta})$ for at least one coordinate q, then*

$$m = O\big(n^{2-\eta/4} + n^{3/2}\big). \tag{6.8}$$

Consequently, if $m = n^{2-o(1)}$, then in every cyclic representation

$$e(\Gamma(A^{(q)})) = n^{3-o(1)}, \qquad \sum_h \binom{\deg_{\Gamma(A^{(q)})} h}{2} = n^{4-o(1)}.$$

Proof. If $m < 2n^{3/2}$, then (6.8) is immediate. Otherwise $S_q \geq m^2/(2n) \geq 2n^2$. Since $H \leq n^2$, we have $S_q^2/H \geq 2S_q$, and (6.7) gives

$$E_q \geq \frac{S_q^2}{4H} \geq \frac{S_q^2}{4n^2} \geq \frac{m^4}{16n^4}.$$

Comparison with $E_q = O(n^{4-\eta})$ gives $m = O(n^{2-\eta/4})$. For the final assertion, (2.1) and (6.4) give $e(\Gamma(A^{(q)})) = S_q/2 = n^{3-o(1)}$, while the preceding lower bound and Theorem 3.3(f) give the wedge estimate. □

7 Recursive pressure overlap

Let a forced hole $h = (i, j)$ have forcing labels $K_h = \{k_1 < \cdots < k_d\}$. Write

$$A(i, c_p) = k_p = A(r_p, j), \qquad P_h = \{(r_p, c_q) : 1 \leq p, q \leq d\}.$$

We call P_h the *pressure grid* of h. By Lemma 3.1, either

$$c_1 < \cdots < c_d < j, \qquad i < r_1 < \cdots < r_d, \tag{7.1}$$

or

$$j < c_1 < \cdots < c_d, \qquad r_1 < \cdots < r_d < i. \tag{7.2}$$

Lemma 7.1 (Triangular pressure bound)**.** *Every forced hole satisfies*

$$|P_h \cap \mathcal{H}(A)| \geq \frac{d(h)(d(h)+1)}{2}. \tag{7.3}$$

Proof. In the orientation (7.1), every cell (r_p, c_q) with $p \leq q$ is a hole. Indeed, a label placed there would be smaller than k_p by row monotonicity and larger than k_q by column monotonicity, which is impossible because $k_p \leq k_q$. In the orientation (7.2), the same argument, with inequalities reversed, shows that all (r_p, c_q) with $p \geq q$ are holes. Either triangular half has $d(d+1)/2$ cells. □

Fix a physical cell $z = (r, c)$, whether filled or empty. Define the signed sets of pressure centres

$$\mathcal{Z}_z^+ = \{(i, j) \in \mathcal{H}(A) : z \in P_{(i,j)},\ i < r,\ c < j\}, \tag{7.4}$$

$$\mathcal{Z}_z^- = \{(i, j) \in \mathcal{H}(A) : z \in P_{(i,j)},\ r < i,\ j < c\}. \tag{7.5}$$

These are the only possibilities, corresponding to (7.1) and (7.2). Put

$$L_0(z) = |\mathcal{Z}_z^+| + |\mathcal{Z}_z^-|, \qquad L_1(z) = \sum_{h : z \in P_h} d(h). \tag{7.6}$$

We use the following finite form of the Loomis–Whitney inequality.

Lemma 7.2 (Projection inequality)**.** *If X is a finite subset of a three-fold Cartesian product, then*

$$|X|^2 \le |\pi_{12}X|\,|\pi_{13}X|\,|\pi_{23}X|.$$

Proof. For $(a,b) \in \pi_{12}X$, let $X_{ab} = \{c : (a,b,c) \in X\}$. Cauchy–Schwarz gives

$$|X|^2 \le |\pi_{12}X| \sum_{a,b} |X_{ab}|^2.$$

The sum counts ordered pairs (a,b,c,c') with $(a,b,c),(a,b,c') \in X$. The map

$$(a,b,c,c') \longmapsto ((a,c),(b,c'))$$

is injective into $\pi_{13}X \times \pi_{23}X$. Hence the sum is at most $|\pi_{13}X|\,|\pi_{23}X|$, proving the result. □

Theorem 7.3 (Recursive pressure support)**.** *For every cell z and each sign $\varepsilon \in \{+,-\}$, the centre set $\mathcal{Z}_z^\varepsilon$ has two canonical injective encodings as pairwise two-comparable subsets of $[n]^3$. If A has m filled cells, then*

$$\boxed{L_0(z) \le 2m, \qquad L_1(z) \le 2m^{3/2}.} \tag{7.7}$$

More precisely, each signed contribution to $L_1(z)$ is at most $m^{3/2}$.

Proof. We prove the assertions for $\mathcal{Z}_z^+$; reversing all inequalities gives the proof for $\mathcal{Z}_z^-$. Let $h = (i,j) \in \mathcal{Z}_z^+$. Since $z = (r,c) \in P_h$, the boundary cells

$$A(r,j) = a_j, \qquad A(i,c) = b_i$$

are filled and both labels belong to K_h. For fixed z, the first label depends only on j and increases strictly with j; the second depends only on i and increases strictly with i.

Because a_j forces h and already occurs in column j, it has a unique occurrence (i,p_{ij}) in row i. Similarly, b_i has a unique occurrence (q_{ij},j) in column j. Consider the two encodings

$$(i,j) \longmapsto (i,j,p_{ij}), \qquad (i,j) \longmapsto (i,j,q_{ij}). \tag{7.8}$$

We verify the first. If two images have the same first coordinate i, then $j < j'$ implies $a_j < a_{j'}$ and row monotonicity gives $p_{ij} < p_{ij'}$. If they have the same second coordinate j, the equal-a_j chain gives $i < i' \Rightarrow p_{ij} < p_{i'j}$. If they have the same third coordinate p, then column monotonicity in column p gives $i < i' \Rightarrow a_j < a_{j'}$, and row monotonicity in row r gives $j < j'$. If all coordinates differ, one of the two triples is smaller in at least two coordinates automatically. Hence the image is pairwise two-comparable. The second encoding follows in the same way, using successively the equal-b_i chain, column monotonicity, and row monotonicity.

The first encoding assigns to (i,j) the filled cell (i,p_{ij}). These cells are distinct: centres in different rows have different first coordinates, while for fixed i the preceding argument gives distinct p_{ij}. Thus $|\mathcal{Z}_z^+| \le m$. The other sign gives $L_0(z) \le 2m$.

For the weighted assertion, form the incidence set

$$\mathcal{I}_z^+ = \{(i,j,k) : (i,j) \in \mathcal{Z}_z^+,\ k \in K_{(i,j)}\}.$$

Its (i,j)-projection is the centre set and has size at most m. Its (i,k)-projection is contained in the set of occupied row–label pairs of A, and its (j,k)-projection is contained in the occupied column–label pairs. Each of the latter sets has size exactly m, since a label occurs at most once in a fixed row or column. Lemma 7.2 therefore gives

$$|\mathcal{I}_z^+|^2 \le |\pi_{ij}\mathcal{I}_z^+|\,|\pi_{ik}\mathcal{I}_z^+|\,|\pi_{jk}\mathcal{I}_z^+| \le m^3.$$

Since $|\mathcal{I}_z^+| = \sum_{h \in \mathcal{Z}_z^+} d(h)$, the positive signed load is at most $m^{3/2}$. The negative signed load obeys the same bound; adding them completes the proof. □

The theorem turns uncontrolled geometric overlap into a recursive incidence problem on genuine two-comparable supports. It also yields an explicit sufficient condition for a power saving.

Corollary 7.4 (Pressure-load criterion)**.** *Let*

$$\Omega_1(A) = \max_{z\in\mathcal{H}(A)} L_1(z), \qquad \Omega_0(A) = \max_{z\in\mathcal{H}(A)} L_0(z).$$

Suppose a class of order-n monotone partial matrices satisfies, uniformly, $\Omega_1(A) \leq Cn^{3-\eta}$ for constants $C < \infty$ and $\eta > 0$. Then every member of the class has

$$m = O_C(n^{2-\eta/6} + n). \tag{7.9}$$

Under the alternative hypothesis $\Omega_0(A) \leq Cn^{2-\eta}$, one has $m = O_C(n^{2-\eta/4} + n)$.

Proof. Write $S = \sum_{h\in\mathcal{H}(A)} d(h) = \sum_k t_k(t_k - 1)$ and $H = n^2 - m$. Double-count the weighted incidences (h, z) with $z \in P_h \cap \mathcal{H}(A)$. Lemma 7.1 and the power-mean inequality give

$$H\Omega_1(A) \geq \frac{1}{2}\sum_h (d(h)^3 + d(h)^2) \geq \frac{S^3}{2H^2}.$$

If $m \geq 2n$, then $S \geq m^2/n - m \geq m^2/(2n)$, and $H \leq n^2$. Consequently

$$m^6 \leq 16n^9\Omega_1(A) \leq 16Cn^{12-\eta},$$

so $m \leq (16C)^{1/6}n^{2-\eta/6}$. The case $m < 2n$ is absorbed by the $O(n)$ term.

For the unweighted count, the same double counting gives

$$H\Omega_0(A) \geq \frac{1}{2}\sum_h (d(h)^2 + d(h)) \geq \frac{S^2}{2H}.$$

When $m \geq 2n$, this yields $m^4 \leq 8n^6\Omega_0(A) \leq 8Cn^{8-\eta}$ and hence the asserted exponent $2 - \eta/4$. ☐

The exponent $3/2$ in Theorem 7.3 is optimal as a function of m.

Proposition 7.5 (Sharp pressure overlap)**.** *For all integers $s, t \geq 1$, put $N = s(s + t) + s + 1$. There is an order-N monotone partial matrix with*

$$m = 2s(s + t + 1)$$

filled cells and a hole z such that

$$L_0(z) \geq s^2, \qquad L_1(z) \geq s^2(t + 2).$$

Taking $t = s$ gives $m = \Theta(N)$ and $L_1(z) = \Theta(N^{3/2}) = \Theta(m^{3/2})$.

Proof. Use rows and columns in $[N]$. Set $c_0 = 1$, $r_0 = N$, and

$$C_{i,u} = 1 + (i - 1)(s + t) + u, \qquad J_j = 1 + s(s + t) + j, \qquad R_{j,u} = s + (j - 1)(s + t) + u.$$

For $i, j \in [s]$ and $u \in [s + t]$, let

$$B_i = i, \qquad K_u = s + u, \qquad A_j = s + t + j = K_{t+j}.$$

Fill

$$A(i, c_0) = B_i, \qquad A(i, C_{i,u}) = K_u,$$
$$A(R_{j,u}, J_j) = u, \qquad A(r_0, J_j) = A_j,$$

and leave every other cell empty. The listed indices lie in $[N]$. Each nonempty row and column is strictly increasing. It remains to check the equal-label chains. A label $u \in [s]$ occurs first at (u, c_0) and then at the cells $(R_{j,u}, J_j)$, which move strictly southeast with j. A label $s+u$ with $u \in [t]$ occurs at $(i, C_{i,u})$ for $i \in [s]$ and then at $(R_{j,s+u}, J_j)$ for $j \in [s]$; both subfamilies are southeast chains, and every cell in the first is northwest of every cell in the second. Finally, a label $s+u$ with $t < u \le s+t$ occurs at $(i, C_{i,u})$ for $i \in [s]$ and once more at (r_0, J_{u-t}). These cells also form a southeast chain. Hence A is monotone.

For every $(i, j) \in [s]^2$, the cell (i, J_j) is a hole forced by exactly

$$\{B_i, A_j, K_1, \ldots, K_t\}.$$

Its pressure grid contains $z = (r_0, c_0)$, using the A_j column arm and the B_i row arm. Therefore these s^2 centres contribute s^2 to $L_0(z)$ and $s^2(t+2)$ to $L_1(z)$. The two displayed families of filled cells are disjoint, each having $s(s+t+1)$ members, which proves the formula for m. Putting $t = s$ gives the stated asymptotics. $\square$

8 Scope and the multiscale obstruction

The results above do not prove (1.1). They instead isolate three limitations of one-scale arguments. First, forcing graphs need not be acyclic. For example,

$$\begin{pmatrix} 1 & 3 & \cdot & \cdot \\ 2 & 4 & \cdot & \cdot \\ \cdot & \cdot & 1 & 2 \\ \cdot & \cdot & 3 & 4 \end{pmatrix}$$

is monotone and its forcing graph contains a four-cycle; the four colours are distinct, as Theorem 3.3 requires. More generally, Proposition 3.5 excludes every strategy based only on forbidding one fixed biclique.

Second, Proposition 7.5 shows that the exponent in the unconditional bound $L_1(z) \le 2m^{3/2}$ cannot be improved using only m. When m is near the quadratic scale, this bound is of cubic order, exactly the scale at which Corollary 7.4 ceases to save a power. Any improvement must therefore use the geometry or density of the recursive centre encodings, not only their cardinality.

Third, the baseline part of the exact slack identity is too small at a slowly subquadratic scale. If formally $m = n^2/L$ with $L \to \infty$ and $L = o(n)$, then $H = (1 - o(1))n^2$ and

$$\frac{1}{2H} \sum_q \left(\frac{m^2}{n} - m \right)^2 = \Theta\!\left(\frac{n^4}{L^4} \right), \qquad \binom{m}{2} = \Theta\!\left(\frac{n^4}{L^2} \right).$$

Thus the exact identity can exclude this scale only if it forces one of the variance, alignment, or unsaturated-inversion terms to be large in a way that persists under restriction. No such cross-scale estimate is proved here.

The recursive pressure theorem supplies a natural carrier for such an estimate: grids through one cell already produce smaller two-comparable sets. A fixed power saving would follow from a density decrement on those supports, from a multiplicative loss on a positive proportion of scales, or from a new interaction among the three cyclic representations. Establishing any one of these remains open.

A Finite verification

The accompanying program `src/verify_finite.py` is a supplementary consistency check; it is not used in any proof. It enumerates the $4^9 = 262144$ arrays in

$$\{\text{empty}, 1, 2, 3\}^{[3]\times[3]},$$

retains exactly the 712 arrays satisfying the three monotonicity conditions, and checks the following statements for every retained array and every cyclic coordinate representation:

(i) common orientation, forcing-graph simplicity, degree equality, chain neighbourhoods, order-preserving colour matchings, rainbow four-cycles, and the edge and wedge identities;

(ii) endpoint exclusion for every row subset of size at least two;

(iii) the saturated-inversion correspondence, cyclic energy packing, the pair partition (6.6), and the exact rational identity (6.3);

(iv) the two pressure-support encodings and all three projection bounds.

The same program verifies Proposition 3.5 for four small parameter pairs and Proposition 7.5 for four small parameter pairs. All variance calculations use exact rational arithmetic. The complete command and expected output are given in the repository README. The finite computation proves only the stated finite enumeration and checks the implementation against the general proofs; the theorems themselves are not computer-assisted.

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